Spatial Estimates of SWE for the Rio Grande Basin

April 8, 2025

Max Gersh[1], Logan Stephenson[2], Ross Palomaki[2], and Karl Rittger[2]
[1] New Mexico Office of the State Engineer (NM OSE) Hydrology Bureau
[2] Institute of Arctic and Alpine Research (INSTAAR), University of Colorado Boulder
Work funded by: NASA Western Water Applications Office (WWAO)
Contact: Max.Gersh@ose.nm.gov

Overview Map – Rio Grande Basin Study Area

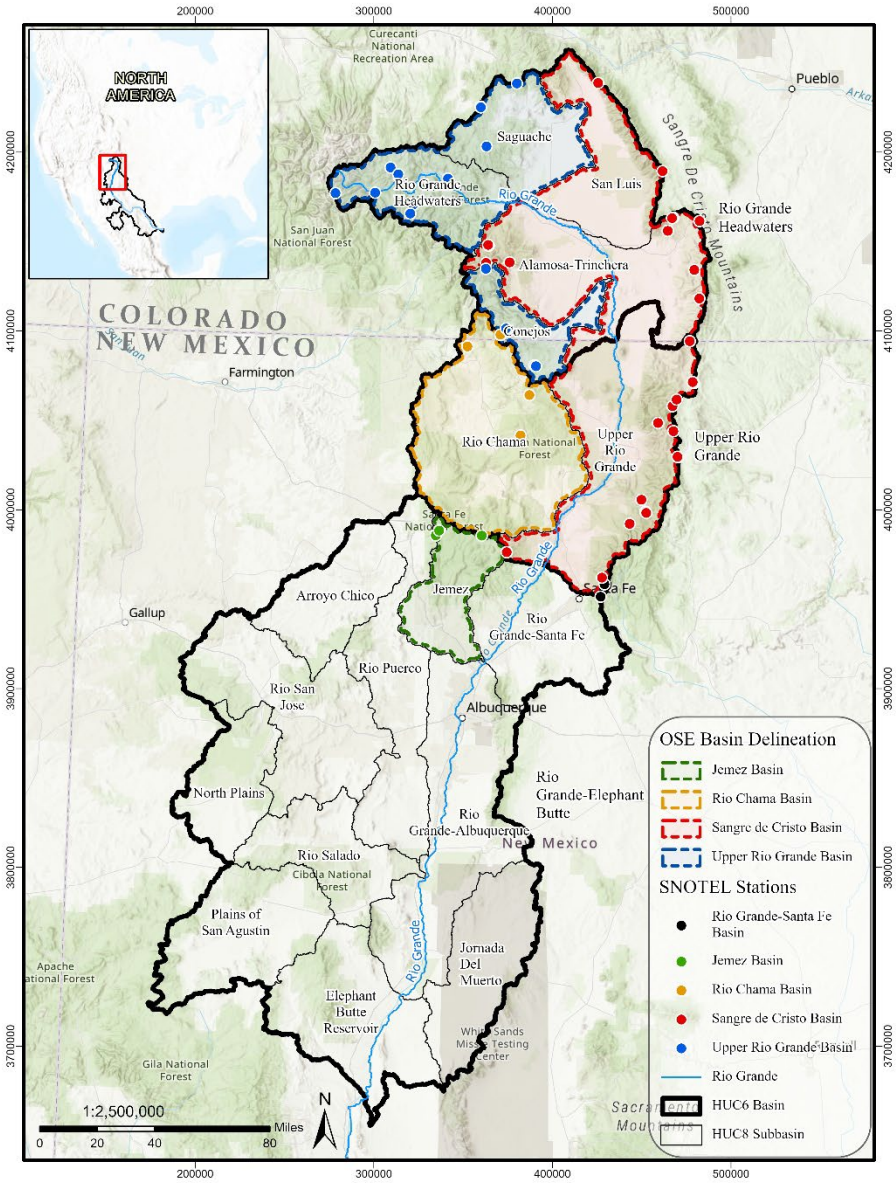


Figure 1. Overview map of the Rio Grande (RG) Basin study area. *The Rio Grande Headwaters, Upper Rio Grande, and Rio Grande-Elephant Butte Basins are presented as the standard HUC6 basins. HUC8 groupings (Jemez – green, Rio Chama – orange, Sangre de Cristo – red, and Upper Rio Grande – blue) are symbolized.*

Summary of SNODAS SWE for April 8, 2025

- Spatial estimates of snow water equivalent (SWE) and change in SWE (presented in inches) for the Rio Grande (RG) Basin study area are presented in Figure 2.
- A storm cycle brought new snow across the study area on April 4-5, resulting in an increase in SWE throughout the RG Basin (Figure 2) since the last report date (April 1, 2025).
- **Table 1**: Estimated SWE amount (median and mean in inches (in)), estimated volume (acre-feet (af)), and change in volume (af) since the last report date (April 1, 2025), are reported for the standard Hydrologic Unit Code 6 (HUC6) basins (Rio Grande Headwaters (Figure 3), Upper Rio Grande (Figure 4), and Rio Grande-Elephant Butte (Figure 5)).
- **Table 2**: Estimated SWE amount (median and mean (in)), estimated volume (af), and change in volume (af) since the last report date (April 1, 2025), are reported for the specific HUC8 subbasin groupings (OSE Basin Delineations shown in Figure 1; Upper Rio Grande, Rio Chama, Sangre de Cristo, and Jemez).
- **Table 3**: Estimated SWE amount (median and mean (in)), estimated volume (af), and change in volume (af) since the last report date (April 1, 2025), are reported by basin and elevation band for the HUC6 basins (Rio Grande Headwaters (Figure 3), Upper Rio Grande (Figure 4), and Rio Grande-Elephant Butte (Figure 5)).
- SWE volume (af) across the Rio Grande Headwaters Basin (HUC6) is largely present throughout the Rio Grande Headwaters subbasin (HUC8) with a majority of that existing between 11,000 and 12,000 feet (Figure 3).
- SWE volume (af) across the Upper Rio Grande Basin (HUC6) is largely present throughout the Rio Chama subbasin (HUC8) with a majority of that existing between 10,000 and 11,000 feet (Figure 4).
- SWE volume (af) across the Rio Grande-Elephant Butte Basin (HUC6) is largely present throughout the Jemez subbasin (HUC8) with a majority of that existing between 8,000 and 9,000 feet (Figure 5).

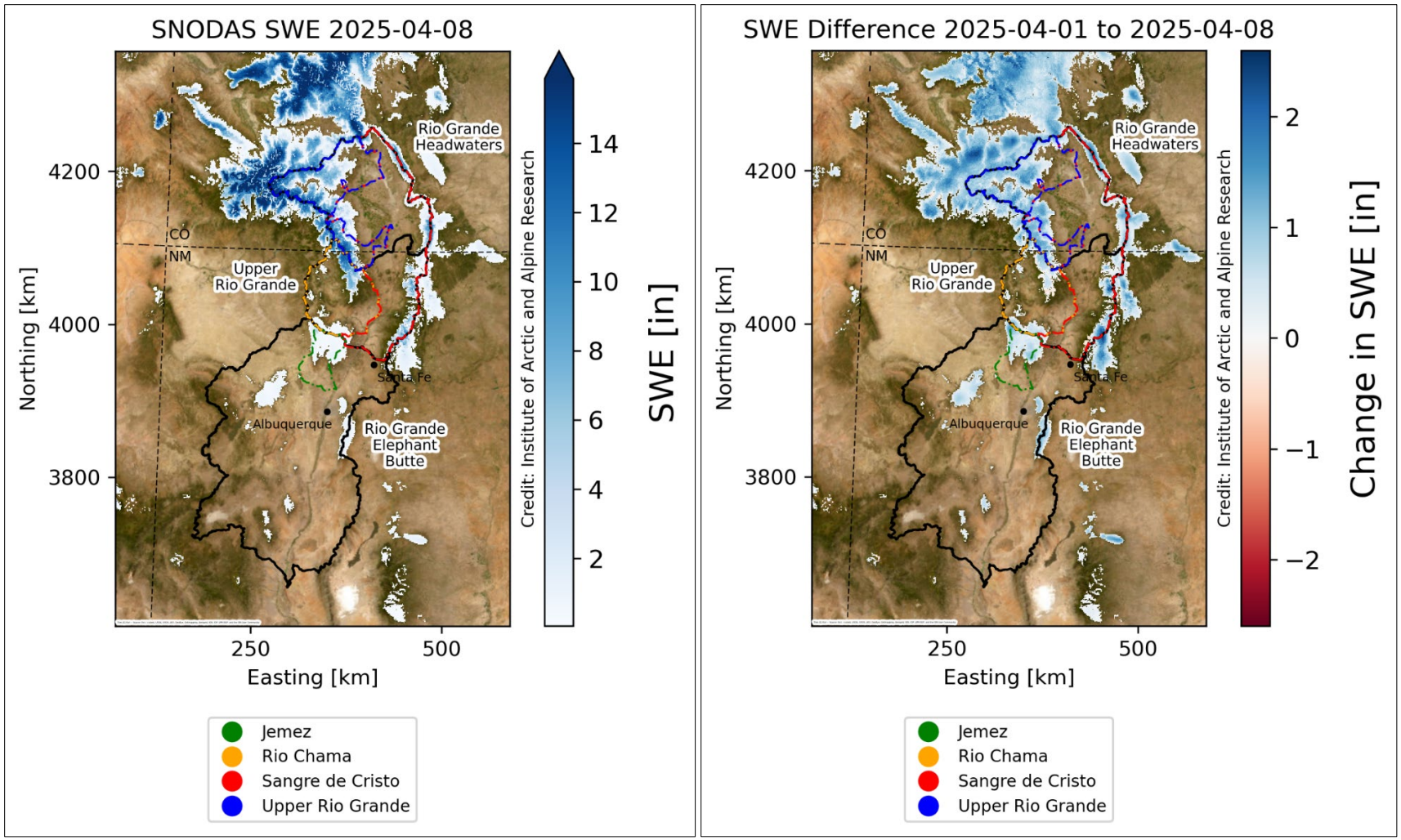


Figure 2. Estimated SNODAS SWE and Change in SNODAS SWE across the Rio Grande Basin study area. *SWE amounts (in) across the RG Basin study area (left); dark blue pixels denote greater SWE amounts. Change in SNODAS SWE amounts (in) across the RG Basin study area from April 1, 2025, to April 8, 2025 (right); blue pixels denote an increase in SWE amounts.*

Introduction

The Rio Grande (RG) is a snowmelt-driven river system with up to 75% of its water originating from seasonal snowpack in the mountains of Colorado and northern New Mexico (Moeser et al., 2021; Figure 2). Snowpack acts as a natural reservoir in the RG Basin, gradually releasing meltwater during the spring and early summer months, sustaining streamflow and providing vital water supply for downstream users. However, regional decreases in snowpack volume and earlier snowmelt timing, especially throughout the western U.S. and New Mexico, can lead to reduced water availability (Mote et al., 2005; EPA, 2016b; Mote et al., 2018; Hale et al., 2023). Long-term changes in snowpack patterns and timing can serve as an indicator of climate change, while understanding short-term changes is critical for monitoring seasonal variability and forecasting seasonal water supply. Monitoring snowpack provides valuable data on shifts in precipitation patterns and temperatures, helping scientists and decision-makers better understand and manage climate-related challenges.

Snow water equivalent (SWE) is the estimate of the depth of liquid water contained in the snowpack. SWE is the standard metric in estimating the volume of meltwater in the snowpack, which plays a critical role in the seasonal snow water supply in the RG Basin. SWE is measured at in-situ Snow Telemetry (SNOTEL) sites throughout the RG Basin and the greater Western U.S. However, the spatial distribution of SWE is complex and often difficult to estimate outside of the in-situ SNOTEL measurements. While SNOTEL provides important point-based measurements, it has been suggested that these data may not be representative of the surrounding snow-covered area (Meromy et al., 2013). Whereas regional tendencies of spatial distribution are consistent, local snowpack volumes are more variable, all of which influences the magnitude and timing of snowmelt-driven runoff.

SWE Analysis Tool

The RG Basin, and more broadly the Western U.S., is experiencing a prolonged and severe drought (EPA, 2016a), exacerbated by a warming climate that increases evapotranspiration, reduces overall snowpack quantity, and reduces the seasonal duration of snowpack. The application of the SWE Analysis Tool (SWEAT) will supply scientists and decision-makers with an improved insight into the spatial distribution of SWE by providing real-time spatial SWE estimates for the RG Basin, its subbasins, and elevation bands within the basin.

Currently, SWEAT incorporates the SNODAS gridded SWE data product (NOHRSC, 2004). Because no spaceborne sensor can directly measure SWE, spatial distribution of SWE must be estimated from remotely sensed observations coupled with modeling approaches. However, gridded SWE products, like SNODAS, are known to have accuracy issues across both mountainous and non-mountainous regions (Mudryk et al., 2025), resulting in a range of spatial SWE estimates for any given location. NM OSE and INSTAAR are working to integrate more data products to build SWEAT based on an ensemble of spatially estimated SWE. By taking an ensemble approach, the goal for SWEAT is to provide spatial SWE estimations for the RG Basin that are, on average, equally or more accurate than any individual model (Thompson, 1977; Arsenault et al., 2015). In addition, the use of a single estimate calculated from an ensemble can reduce confusion for water managers about which product may be performing best, providing a path towards consistency throughout users, and is useful in identifying product outliers.

SNODAS

The SNOw Data Assimilation System (SNODAS) represents a modeling and data assimilation system developed by the National Operational Hydrologic Remote Sensing Center (NOHRSC). SNODAS generates spatial estimates of SWE, snow cover, and snow depth through a combination of remotely sensed snow data and in-situ measurements of SWE from SNOTEL at a 1-km spatial resolution and 24-hour temporal resolution.

The national SNODAS dataset is available through the SNODAS Data Products at National Snow and Ice Data Center (NSIDC), Version 1 (NOHRSC, 2004). SNODAS data can be viewed at the NOHRSC Interactive Snow Information page. Gridded observed snowfall images, seasonal totals, and data downloads in several formats can be found on the National Gridded Snowfall Analysis page.

Refer to the SNODAS User Guide and the NSIDC Special Report 11 (Barrett, 2003) for more information.

Discussion

Figures and tables for the identified basins (Figure 1) are presented below. The median and mean SWE (in) is presented to avoid potentially skewed outliers, most notably due to an uneven distribution of SWE across elevation bands. While the mean value represents the average value, it is highly influenced by outliers and is typically best used for uniform distributions of data. In contrast, the median value represents the middle value and is less sensitive to outliers, which can be more useful in skewed distributions. Note that the basin-wide median and mean values may not be directly comparable with the SNOTEL basin-wide values. If comparisons are made with SNOTEL sites, a better comparison may be made across elevation bands (Table 3).

Figure 3 (Rio Grande Headwaters), Figure 4 (Upper Rio Grande), and Figure 5 (Rio Grande-Elephant Butte) show the spatial distribution of SWE amount (in) and SWE volume (af) across the HUC6 basins. Since the last report date (April 1, 2025), a storm cycle brought new snow across the study area, resulting in an increase in SWE.

Estimated SWE amounts (in), volumes (af), and change in volumes (af) since the last report are presented in the tables below.

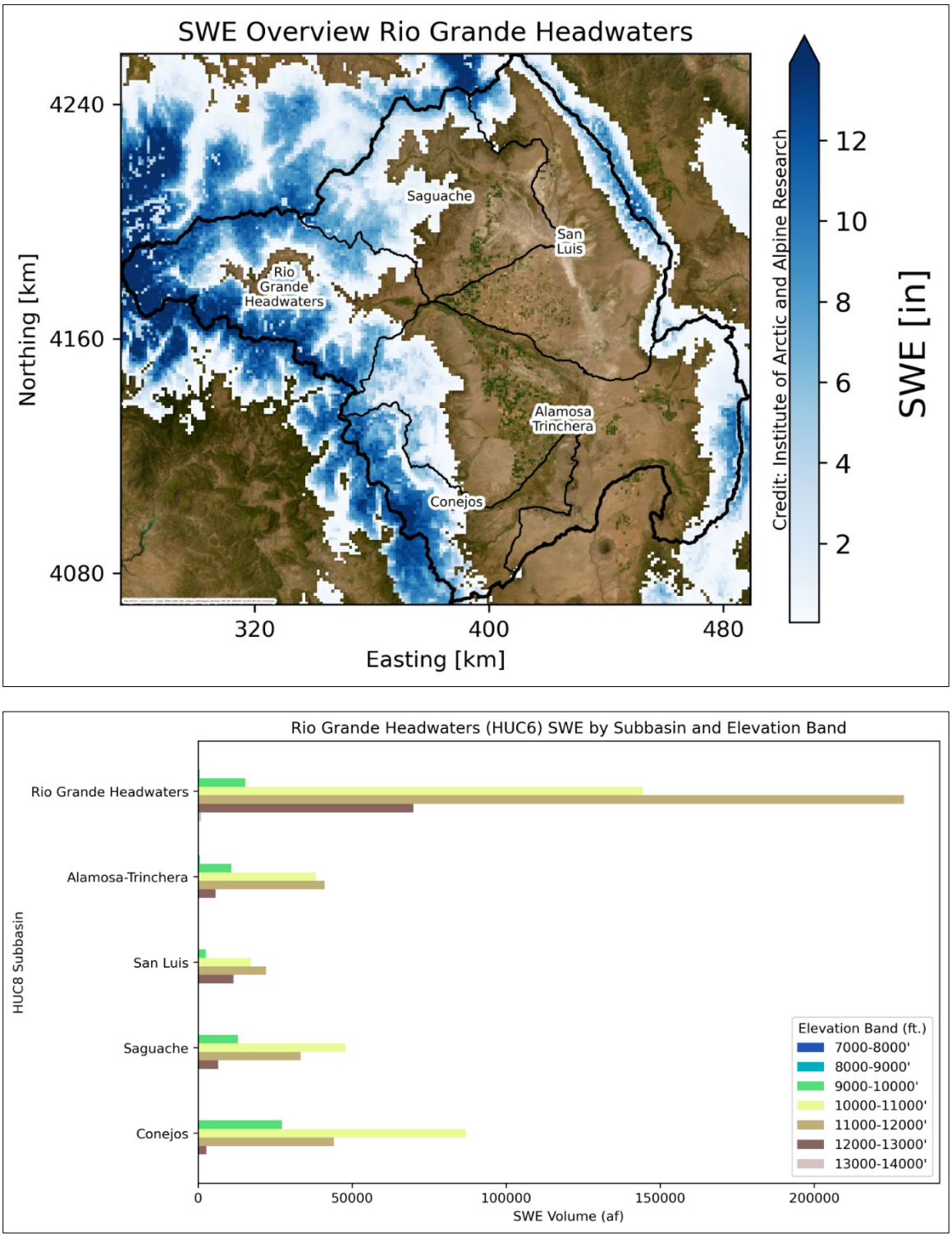
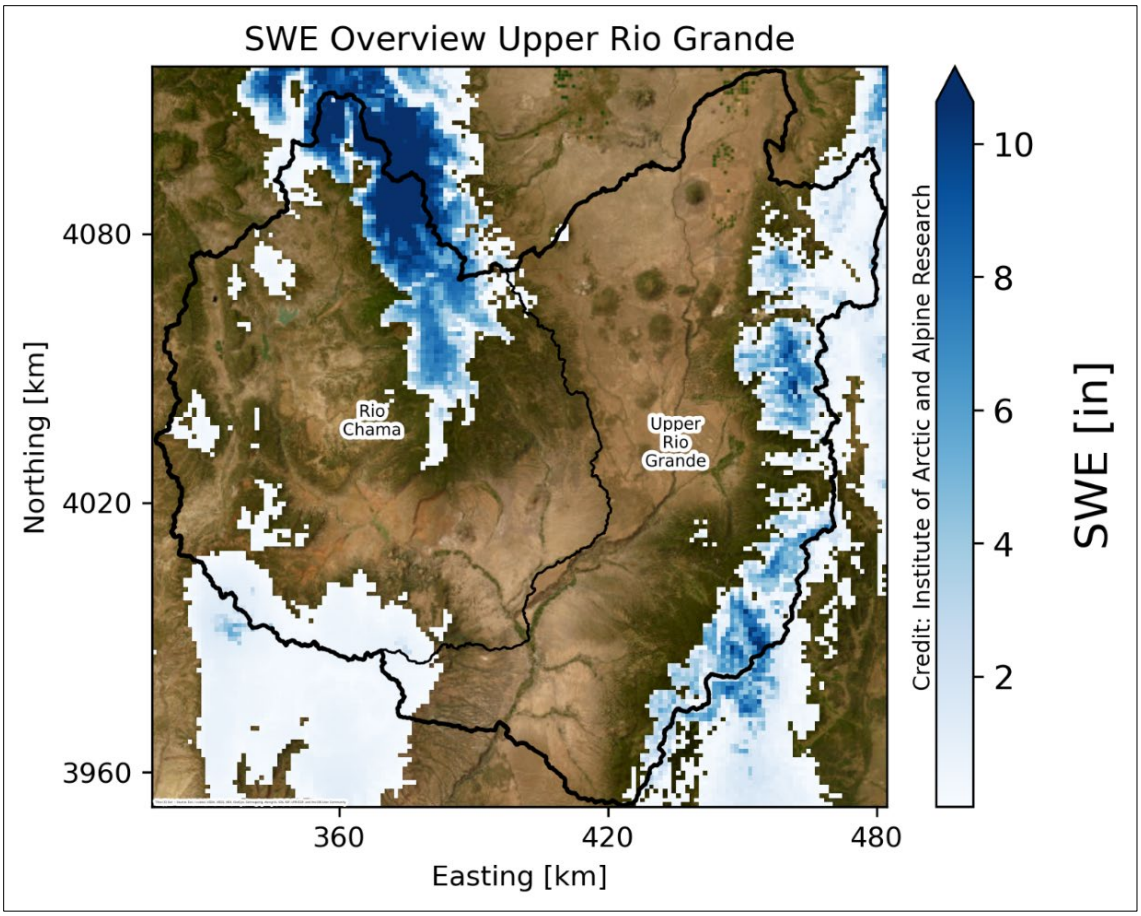


Figure 1. Estimated SNODAS SWE (in) across the Rio Grande Headwaters Basin for April 8, 2025. *SWE amount (in) across the Rio Grande Headwaters Basin (HUC6); including the Rio Grande Headwaters, Alamosa-Trinchera, San Luis, Saguache, and Conejos subbasins (top). SWE volume (af) by subbasin and elevation band across the Rio Grande Headwaters Basin (bottom).*



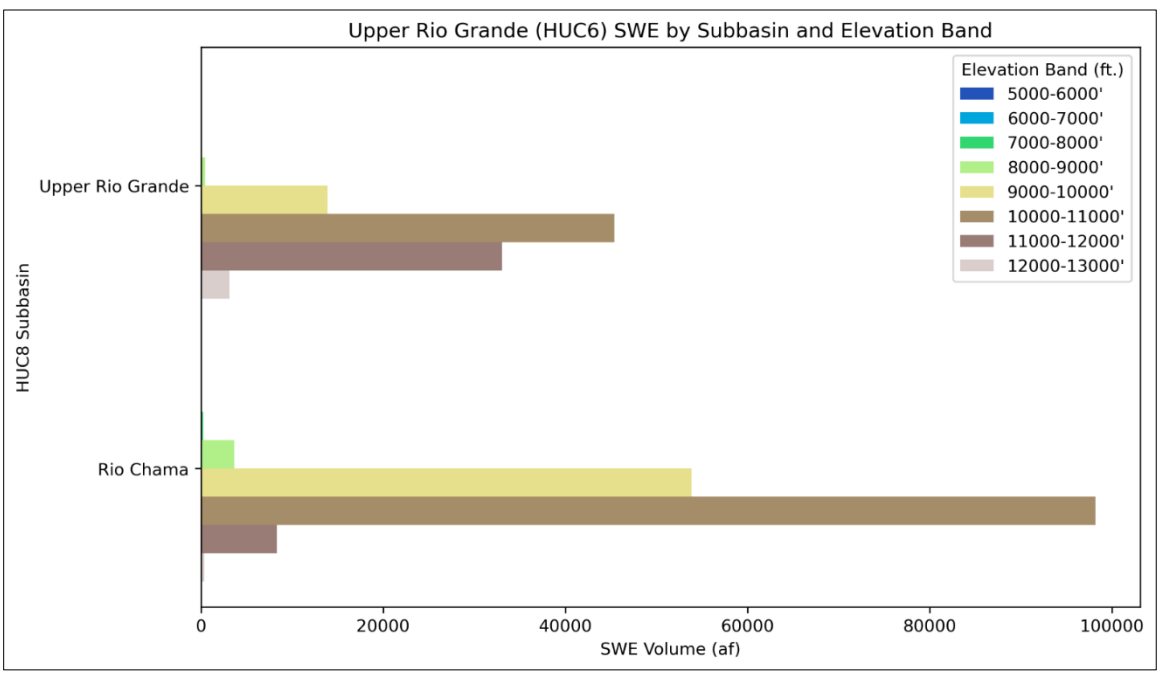
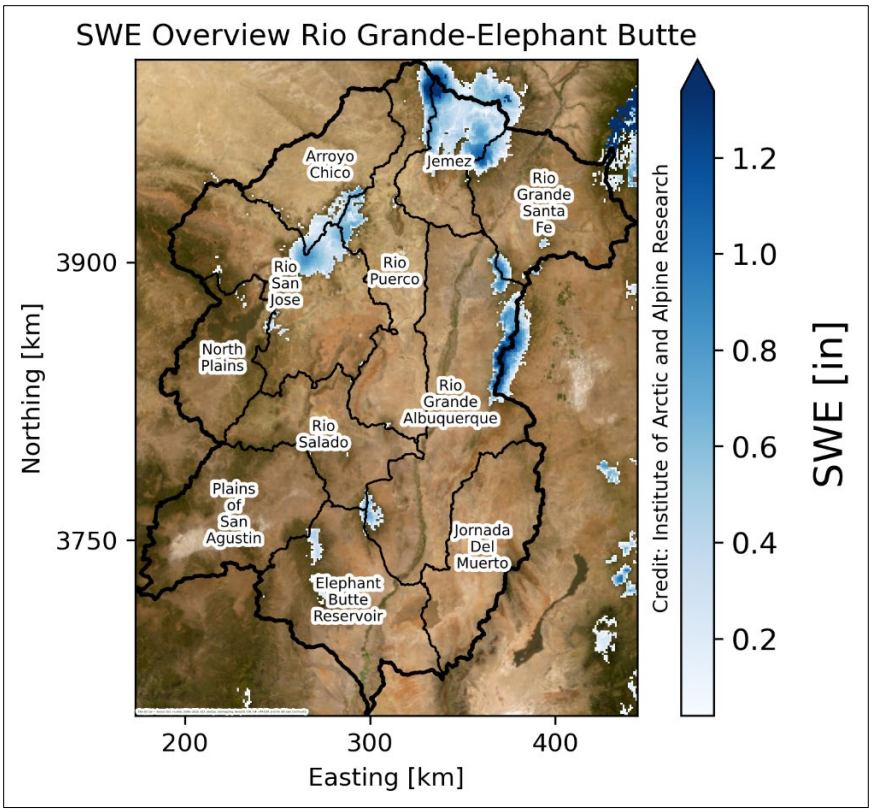


Figure 2. Estimated SNODAS SWE (in) across the Upper Rio Grande Basin for April 8, 2025. *SWE amount (in) across the Upper Rio Grande Basin (HUC6); including the Upper Rio Grande and Rio Chama subbasins (top). SWE volume (af) by subbasin and elevation band across the Upper Rio Grande Basin (bottom).*



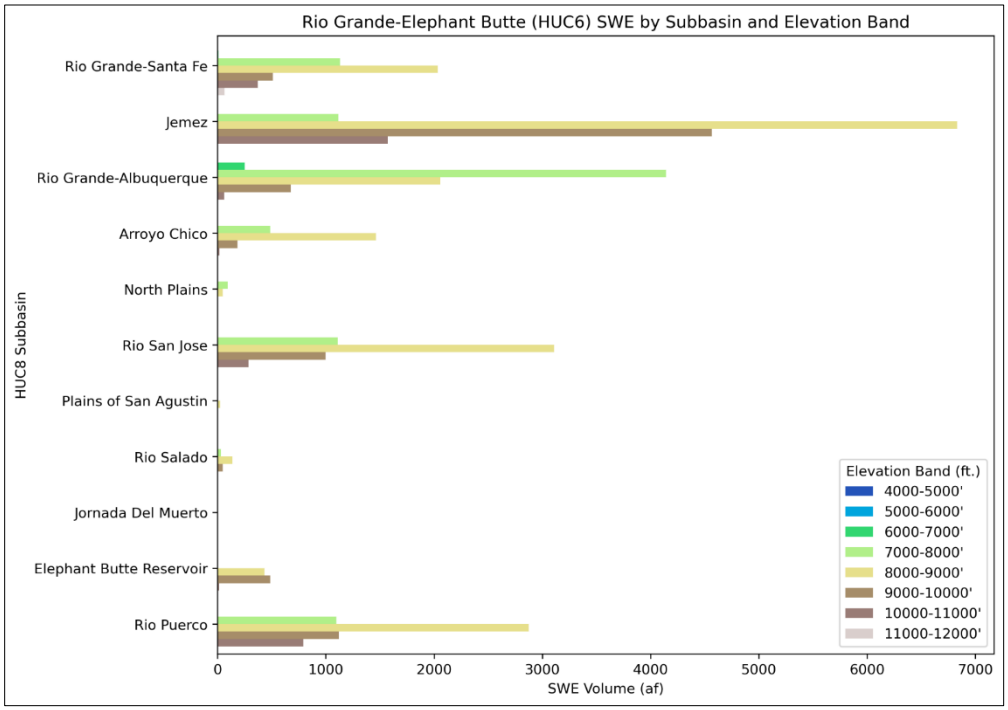


Figure 3. Estimated SNODAS SWE (in) across the Rio Grande-Elephant Butte Basin for April 8, 2025. *SWE amount (in) across Rio Grande-Elephant Butte (HUC6); including the Rio Grande-Santa Fe, Jemez, Rio Grande-Albuquerque, Arroyo Chico, North Plains, Rio San Jose, Plains of San Agustin, Rio Salado, Jornada Del Muerto, Elephant Butte Reservoir, and Rio Puerco subbasins (top). SWE volume (af) by subbasin and elevation band across the Rio Grande-Elephant Butte Basin (bottom).*

Tables

Table 1. Estimated SWE by HUC6 Basin for April 8, 2025. *Shown are median SWE (in.), mean SWE (in.), estimated volume (af), and volume change (af) since the last report date. Values are separated by HUC8 subbasin and totaled for each HUC8 basin (bold).*

HUC6 Basin	HUC8 Subbasin	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 1, 2025 (af)
Rio Grande Headwaters	Rio Grande Headwaters	6.4	6.3	460,261	54,706
	Alamosa-Trinchera	0.0	0.7	96,407	27,491
	San Luis	0.0	0.6	53,339	10,890
	Saguache	0.0	1.4	100,714	20,221
	Conejos	0.5	3.9	161,191	19,900
	Total	**0.0**	**2.1**	**871,912**	**133,209**
Upper Rio Grande	Upper Rio Grande	0.0	0.6	95,903	20,696
	Rio Chama	0.0	1.0	164,623	15,711
	Total	**0.0**	**0.8**	**260,526**	**36,408**
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	0.0	0.0	4,122	3,702
	Jemez	0.0	0.3	14,083	12,482
	Rio Grande-Albuquerque	0.0	0.0	7,187	7,187
	Arroyo Chico	0.0	0.0	2,152	2,152
	North Plains	0.0	0.0	142	142
	Rio San Jose	0.0	0.0	5,499	5,499
	Plains of San Agustin	0.0	0.0	24	24
	Rio Salado	0.0	0.0	218	218
	Jornada Del Muerto	0.0	0.0	0	0
	Elephant Butte Reservoir	0.0	0.0	931	931
	Rio Puerco	0.0	0.1	5,887	5,692
	Total	**0.0**	**0.0**	**40,247**	**38,030**

Table 2. Estimated SWE by Specified Subbasin Groupings for April 8, 2025. *Shown are median SWE (in.), mean SWE (in.), estimated volume (af), and volume change (af) since the last report date. Values are separated by HUC8 subbasin and totaled for each specified subbasin grouping (bold).*

Subbasin Grouping	HUC8 Subbasin	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 1, 2025 (af)
Upper Rio Grande	Rio Grande Headwaters	6.4	6.3	460,261	54,706
	Saguache	0.0	1.4	100,714	20,221
	Conejos	0.5	3.9	161,191	19,900
	Total	**0.8**	**3.9**	**720,339**	**94,384**
Sangre de Cristo	San Luis	0.0	0.6	53,339	10,890
	Alamosa-Trinchera	0.0	0.7	96,407	27,491
	Upper Rio Grande	0.0	0.6	95,903	20,696
	Total	**0.0**	**0.6**	**244,575**	**59,104**
Rio Chama		**0.0**	**1.0**	**165,743**	**15,838**
Jemez		**0.0**	**0.3**	**14,337**	**12,736**

Table 3. Estimated SWE by HUC6 Basin and Elevation Band for April 8, 2025. *Shown are median SWE (in.), mean SWE (in.), estimated volume (af), and volume change (af) since the last report date. Values are separated by elevation band (1000 ft. increments) and HUC8 subbasin.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 1, 2025 (af)
Rio Grande Headwaters	Rio Grande Headwaters	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	441	441
		9000-10000'	0.4	1.1	15,349	5,484
		10000-11000'	6.5	6.6	144,425	19,306
		11000-12000'	10.6	10.5	229,149	23,377
		12000-13000'	12.2	11.7	69,875	5,980
		13000-14000'	10.6	8.3	1,021	119
	Alamosa-Trinchera	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	537	537
		9000-10000'	0.3	0.5	10,799	8,499
		10000-11000'	2.4	3.2	38,178	9,351
		11000-12000'	4.4	4.8	41,081	7,459
		12000-13000'	2.1	3.1	5,715	1,579
		13000-14000'	1.2	1.2	97	65
	San Luis	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	16	16
		9000-10000'	0.0	0.3	2,539	1,088
		10000-11000'	1.5	2.9	17,191	3,287
		11000-12000'	4.2	4.7	22,017	3,662
		12000-13000'	3.5	4.4	11,497	2,796
		13000-14000'	0.5	1.3	79	41
	Saguache	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	41	-23
		9000-10000'	0.5	0.8	12,925	4,171
		10000-11000'	3.7	3.8	47,784	9,634
		11000-12000'	7.4	7.2	33,372	5,316
		12000-13000'	7.0	7.1	6,546	1,103
		13000-14000'	2.2	2.2	45	20
	Conejos	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	240	189
		9000-10000'	0.5	2.5	27,222	3,709
		10000-11000'	9.8	9.2	86,900	9,590

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 1, 2025 (af)
	Conejos	11000-12000'	9.6	9.0	44,096	5,942
		12000-13000'	7.0	7.0	2,733	470
Upper Rio Grande	Upper Rio Grande	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	477	328
		9000-10000'	0.1	0.6	13,911	4,659
		10000-11000'	2.1	2.7	45,360	7,436
		11000-12000'	4.1	4.2	33,047	6,880
		12000-13000'	2.3	2.6	3,109	1,394
	Rio Chama	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	231	231
		8000-9000'	0.0	0.1	3,628	2,508
		9000-10000'	0.9	2.6	53,849	4,055
		10000-11000'	9.0	8.8	98,192	7,994
		11000-12000'	10.6	10.4	8,343	874
		12000-13000'	9.2	9.2	379	49
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	9	9
		7000-8000'	0.0	0.1	1,132	1,132
		8000-9000'	0.4	0.4	2,035	2,035
		9000-10000'	0.4	0.4	511	511
		10000-11000'	1.5	1.5	374	21
		11000-12000'	0.5	0.8	62	-5
	Jemez	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	1	1
		7000-8000'	0.2	0.2	1,116	1,116
		8000-9000'	0.4	0.5	6,831	6,831
		9000-10000'	0.6	0.8	4,563	3,531
		10000-11000'	1.3	1.6	1,571	1,002
	Rio Grande-Albuquerque	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	251	251
		7000-8000'	0.3	0.4	4,141	4,141

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 1, 2025 (af)
Rio Grande-Elephant Butte	Rio Grande-Albuquerque	8000-9000'	0.6	0.7	2,057	2,057
		9000-10000'	0.8	0.8	676	676
		10000-11000'	0.8	0.7	62	62
	Arroyo Chico	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	487	487
		8000-9000'	0.3	0.3	1,465	1,465
		9000-10000'	0.7	0.7	184	184
		10000-11000'	0.8	0.8	16	16
	North Plains	6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	95	95
		8000-9000'	0.0	0.0	47	47
	Rio San Jose	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	1	1
		7000-8000'	0.0	0.0	1,108	1,108
		8000-9000'	0.0	0.2	3,107	3,107
		9000-10000'	0.7	0.7	998	998
		10000-11000'	0.8	0.8	285	285
	Plains of San Agustin	6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	24	24
		9000-10000'	0.0	0.0	0	0
	Rio Salado	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	32	32
		8000-9000'	0.0	0.1	137	137
		9000-10000'	0.6	0.5	49	49
	Jornada Del Muerto	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
	Elephant Butte Reservoir	4000-5000'	0.0	0.0	0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 1, 2025 (af)
Rio Grande-Elephant Butte	Elephant Butte Reservoir	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.1	433	433
		9000-10000'	0.3	0.3	486	486
		10000-11000'	0.6	0.6	13	13
	Rio Puerco	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	3	3
		7000-8000'	0.0	0.1	1,097	1,097
		8000-9000'	0.6	0.6	2,876	2,876
		9000-10000'	1.1	1.1	1,120	1,060
		10000-11000'	1.3	1.5	791	656

Acknowledgments

The work completed and presented in this report is funded by NASA Western Water Applications Office (WWAO), subcontract no. 1712706, as part of the *Rio Grande River Basin Needs Assessment Report* (WWAO and DBS&A, 2022).

References

Arsenault, R., Gatien, P., Renaud, B., Brissette, F., & Martel, J. L. (2015). A comparative analysis of 9 multi-model averaging approaches in hydrological continuous streamflow simulation. *Journal of Hydrology*, 529, 754-767.

Barrett, Andrew. (2003). *National Operational Hydrologic Remote Sensing Center Snow Data Assimilation System (SNODAS) Products at NSIDC*. NSIDC Special Report 11. Boulder, CO USA: National Snow and Ice Data Center. 19 pp.

Environmental Protection Agency [EPA]. (2016a). *Climate Change Indicators: Drought*. Web update: June 2024. Available at: https://www.epa.gov/climate-indicators/climate-change-indicators-drought. Accessed: March 31, 2025.

Environmental Protection Agency [EPA]. (2016b) *Climate Change Indicators: Snowpack*. Web update: June 2024. Available at: https://www.epa.gov/climate-indicators/climate-change-indicators-snowpack. Accessed: March 31, 2025.

Hale, K.E., Jennings, K.S., Musselman, K.N., Livneh, B., Molotch, N.P. (2023). Recent decreases in snow water storage in western North America. *Communications Earth & Environment*, 4(1), p. 170. Available at: https://doi.org/10.1038/s43247-023-00751-3.

Meromy, L., Molotch, N.P., Link, T.E., Fassnacht, S.R., & Rice, R. (2013). Subgrid variability of snow water equivalent at operational snow stations in the western USA. *Hydrological Processes, 27*, 2383-2400, doi: 10.1002/hyp.9355.

Moeser, C. David., Chavarria, S. B., & Wootten, A. M. (2021). *Streamflow response to potential changes in climate in the Upper Rio Grande Basin* (USGS Numbered Series No. 2021–5138; Scientific Investigations Report). U.S. Geological Survey. https://www.usgs.gov/publications/streamflow-response-potential-changes-climate-upper-rio-grande-basin.

Mote, P. W., Hamlet, A. F., Clark, M. P., & Lettenmaier, D. P. (2005). Declining mountain snowpack in western North America. *Bulletin of the American Meteorological Society, 86*(1), 39–50. https://doi.org/10.1175/BAMS-86-1-39.

Mote, P. W., Li, S., Lettenmaier, D. P., Xiao, M., & Engel, R. (2018). Dramatic declines in snowpack in the western US. *Climate and Atmospheric Science*, 1, 2. https://doi.org/10.1038/s41612-018-0012-1.

Mudryk, L., Mortimer, C., Derksen, C., Elias Chereque, A., Kushner, P. (2025). Benchmarking of snow water equivalent (SWE) products based on outcomes of the SnowPEx+ Intercomparison Project. *The Cryosphere, 19*(1), pp. 201–218. Available at: https://doi.org/10.5194/tc-19-201-2025.

NASA Wester Water Applications Office [WWAO] and Daniel B. Stephens & Associates, Inc. [DBS&A]. (2022). *Rio Grande River Basin Needs Assessment Report.* NASA WWAO Assessment Report. Available at: https://wwao.jpl.nasa.gov/documents/138/Water-Needs-Rio-Grande-Report-2022.pdf.

National Operational Hydrologic Remote Sensing Center [NOHRSC]. (2004). *Snow Data Assimilation System (SNODAS) Data Products at NSIDC, Version 1*. SWE Dataset. Boulder, Colorado USA. NSIDC: National Snow and Ice Data Center. https://doi.org/10.7265/N5TB14TC. Accessed: April 1, 2025, and as needed.

Thompson, P.D. (1977). How to improve accuracy by combining independent forecasts. *Monthly Weather Review, 105*(2), pp.228-229.